

# DATA VISUALISATION

The background is a solid teal color. On the left side, there is a complex network of white lines connecting various white dots, forming a web-like structure. Scattered across the background are several faint, light blue geometric shapes, including triangles and polygons. On the right side, there are several circular icons representing different data visualization and technology concepts: a bar chart, a pie chart, a line graph, a magnifying glass over a globe, a 24-hour clock, a document with a pencil, a folder with an 'i' icon, a laptop with a gear icon, and a mobile phone icon. These icons are arranged in a way that suggests a flow or relationship between different data analysis tools and concepts.

# What is Data Visualisation?

Data visualization is the **representation of data** or information in a **graph, chart, or other visual format**.

It **communicates relationships of the data with images**.

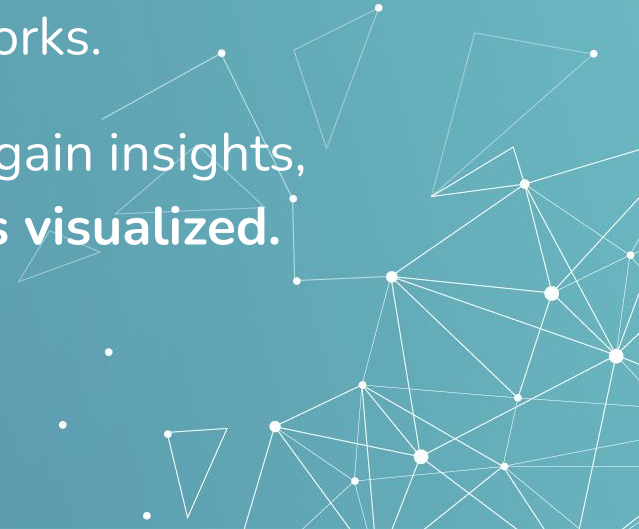


# Why Data Visualisation?

A visual summary of information makes it **easier to identify patterns, trends and outliers** than looking through **thousands of rows on a spreadsheet**.

It's the way the human brain works.

Since the purpose of data analysis is to gain insights, **data is much more valuable when it is visualized.**



# ***“A picture is worth a thousand words”***

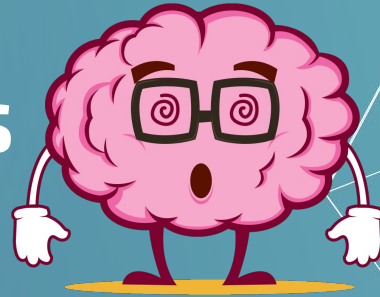
Our **eyes are drawn to colors and patterns**. We can quickly identify red from blue, square from circle. Our culture is visual, including everything from art and advertisements to TV and movies

Data visualization is another form of visual art that **grabs our interest and keeps our eyes on the message.**

An abstract geometric pattern consisting of white lines and dots of varying sizes, forming a network-like structure that extends from the bottom right corner towards the center of the slide. The lines connect the dots, creating a series of interconnected triangles and polygons. The dots are scattered throughout, with some acting as central nodes and others as peripheral points. The overall effect is a complex, organic-looking network that contrasts with the solid teal background.

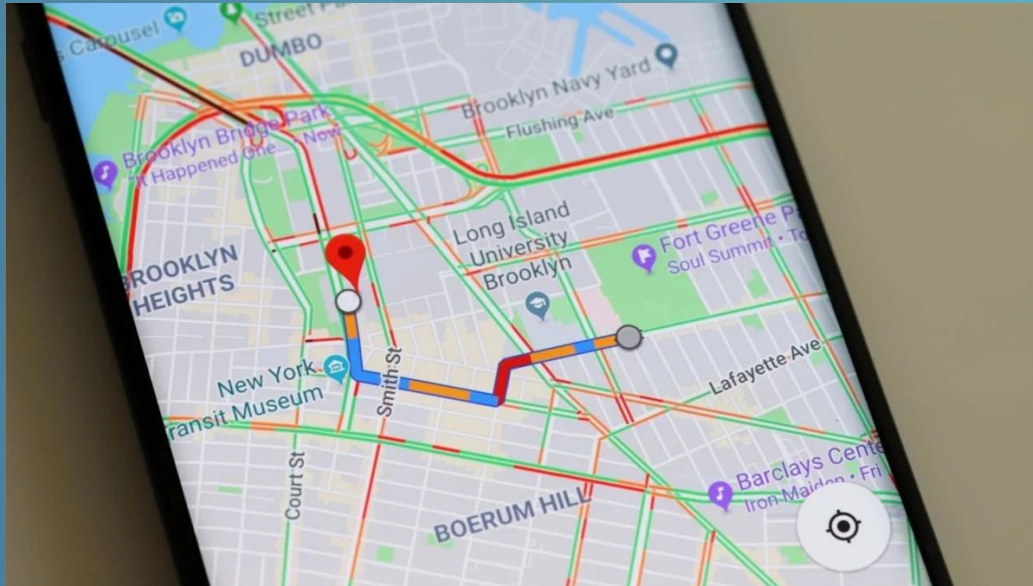
**You've been  
surrounded by  
Data Visualisations  
all this while!**

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# Google Map Traffic visualization

The live traffic status in Google map is another simple example of what is possible through data visualization to make humans life simple and easy. You can see the **red colour indicating slow traffic area** in the city on the map and **blue colour indicating fast** etc.





# Hard Disk Drives interactive Data Visualization

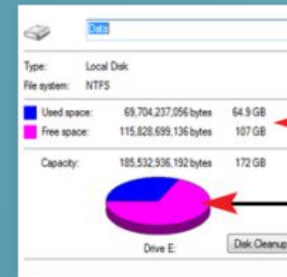
The **blue line and empty space in horizontal bar** and in **pie chart** is an example to **inform about the consumed and free space in hard disk**.

Hard drives data status in interactive visualization is **effective to understand even for a normal user**.

And that's always the target of data visualization to make it so much easy and relevant for people to understand and take quick decision.



Hard disk data status  
in interactive visualization



Hard disk data status  
in interactive Pie Chart visualization



# Geo Spatial Data Visualization



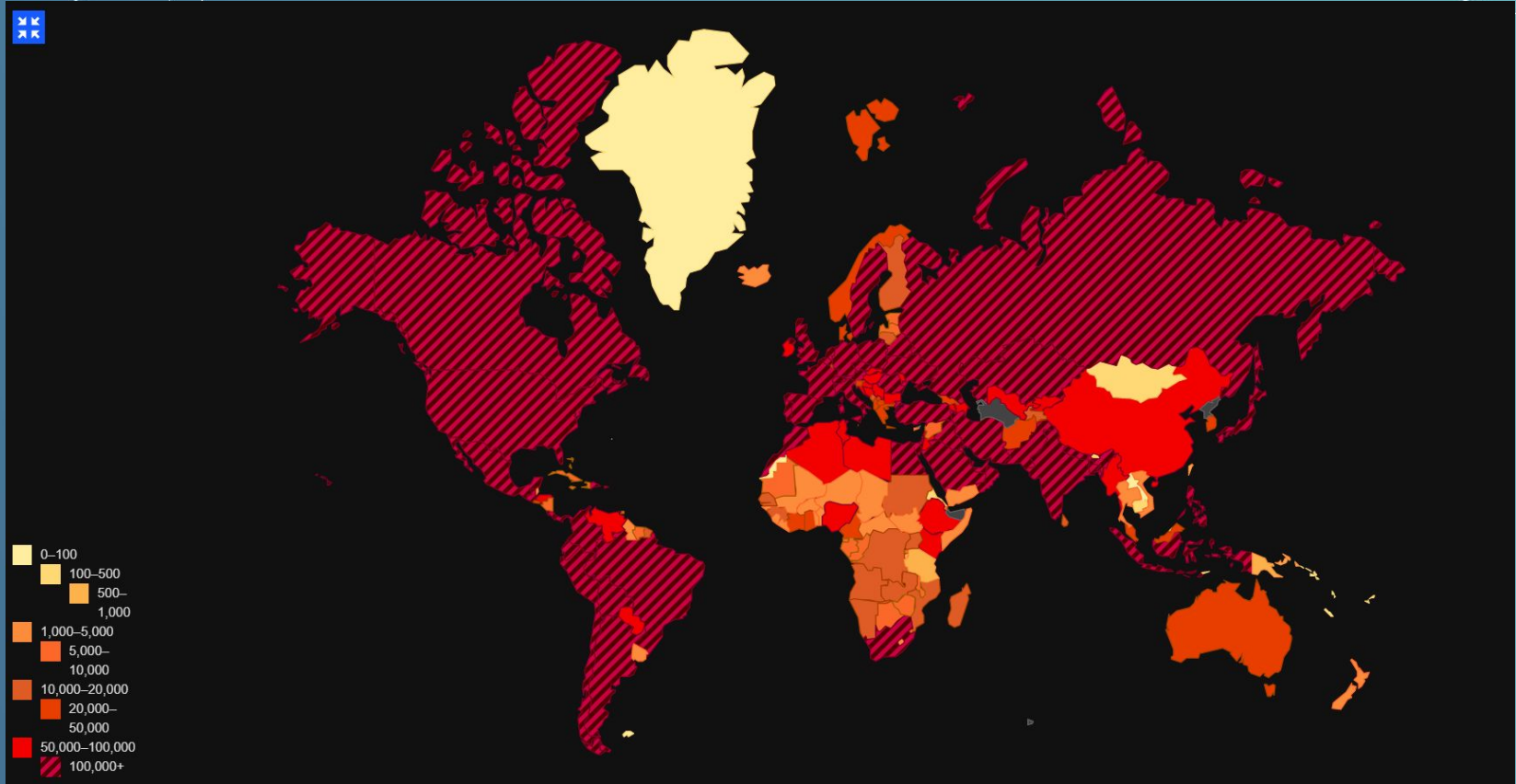
Geospatial visualizations are one of the earliest forms of information visualizations.

Amongst the various types of geospatial visualisations, **Heat maps** are useful when you have to represent large sets of continuous data on a map using a color spectrum.

The map of the world in the next slide shows the world population affected with covid using different shades of red. **The darker the shade of red, the higher the affected population.**



# Geo Spatial Data Visualization

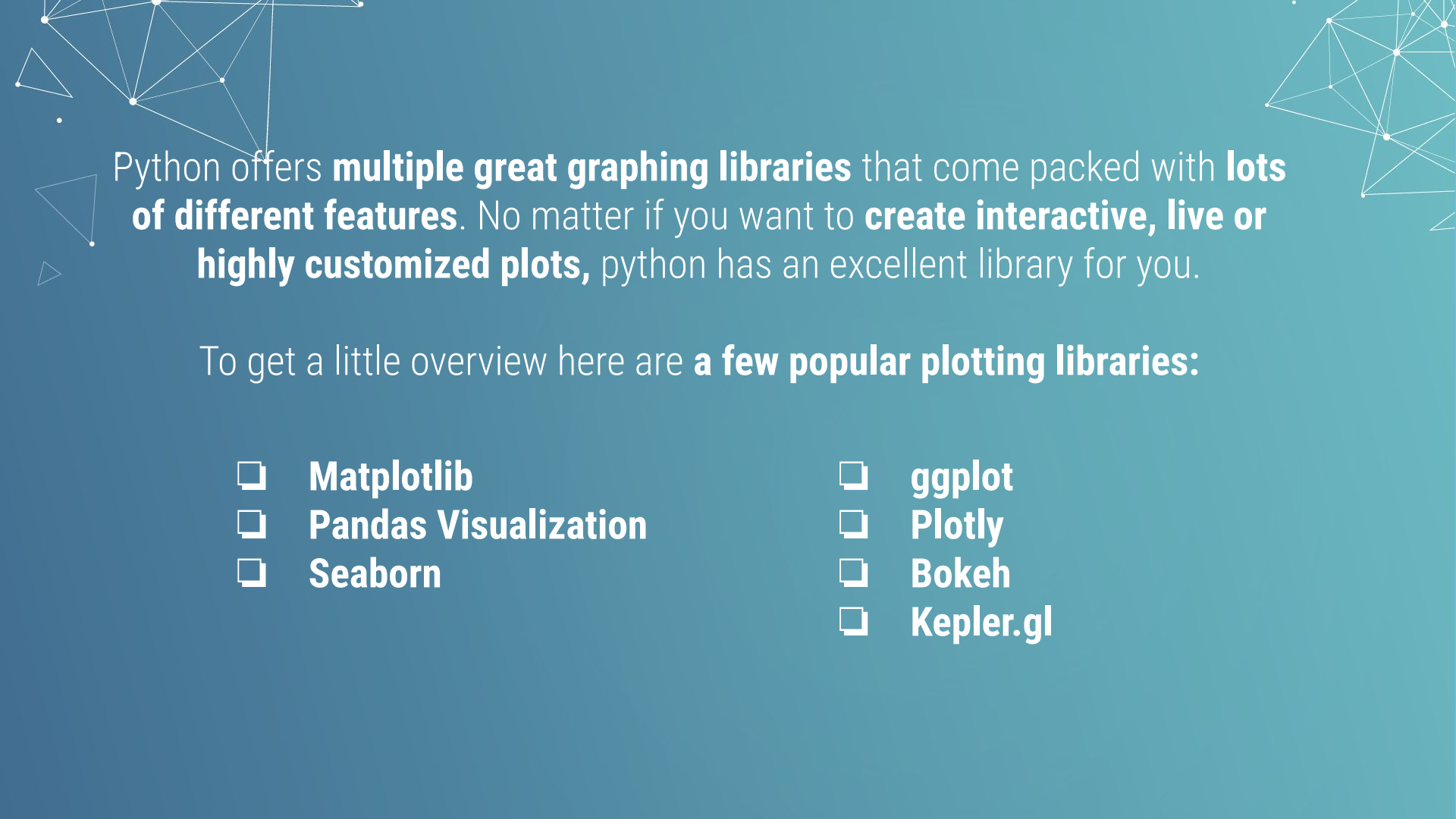


Source: wuflu.live

Can you think of more such examples?

The background of the slide is a solid teal-blue color. Overlaid on this background is a complex, abstract geometric pattern. This pattern consists of numerous small white dots (nodes) connected by thin white lines (edges). These connections form a variety of shapes, including triangles, quadrilaterals, and larger, more irregular polygons. Some areas are more densely connected, while others are more sparse. The overall effect is a network-like or molecular structure. The text 'Data Visualisation in Python' is positioned in the lower right quadrant of the image, rendered in a clean, white, sans-serif font. A thin white horizontal line is located directly beneath the word 'Python'.

# Data Visualisation in Python



Python offers **multiple great graphing libraries** that come packed with **lots of different features**. No matter if you want to **create interactive, live or highly customized plots**, python has an excellent library for you.

To get a little overview here are **a few popular plotting libraries**:

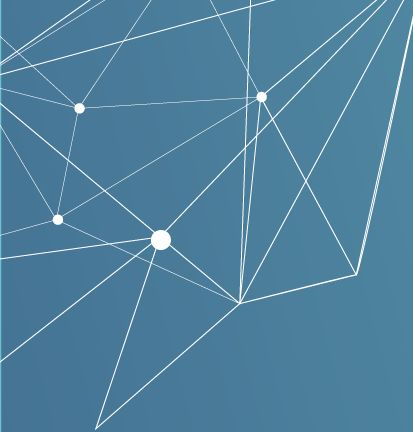
- ❏ **Matplotlib**
- ❏ **Pandas Visualization**
- ❏ **Seaborn**

- ❏ **ggplot**
- ❏ **Plotly**
- ❏ **Bokeh**
- ❏ **Kepler.gl**



**Matplotlib**

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Matplotlib is the **most popular  
python plotting library**.

It is a **low-level library** with a  
Matlab like interface which offers  
lots of freedom at the cost of  
having to write more code.

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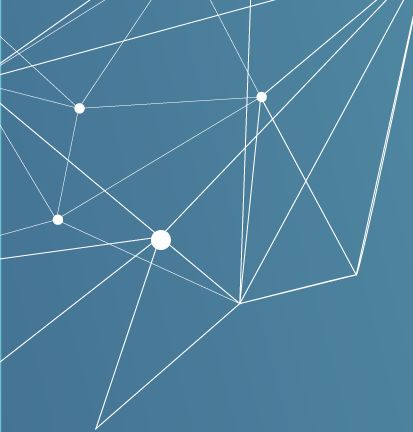




**Seaborn**

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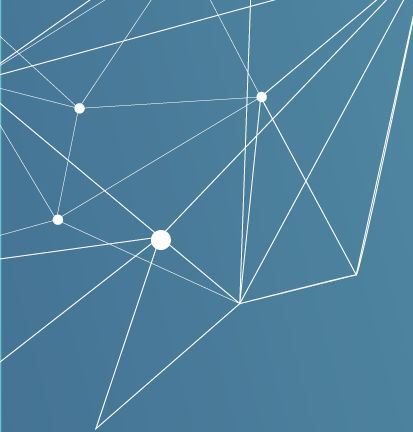
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Seaborn is a **graphic library**  
**built on top of Matplotlib.**

It allows to **make your charts**  
**prettier**, and facilitates some of  
the common data visualisation needs.

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**Let's see how to work  
with these two data  
visualisation libraries**





**Dataset we'll be  
working on**

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# Standard\_Metropolitan\_Areas\_Data

It contains **data of 99 standard metropolitan areas in the US**. The data set provides information on **10 variables** for each area for the period 1976-1977. The areas have been divided into 4 geographic regions: 1=North-East, 2=North-Central, 3=South, 4=West. The variables provided are listed in the table below:

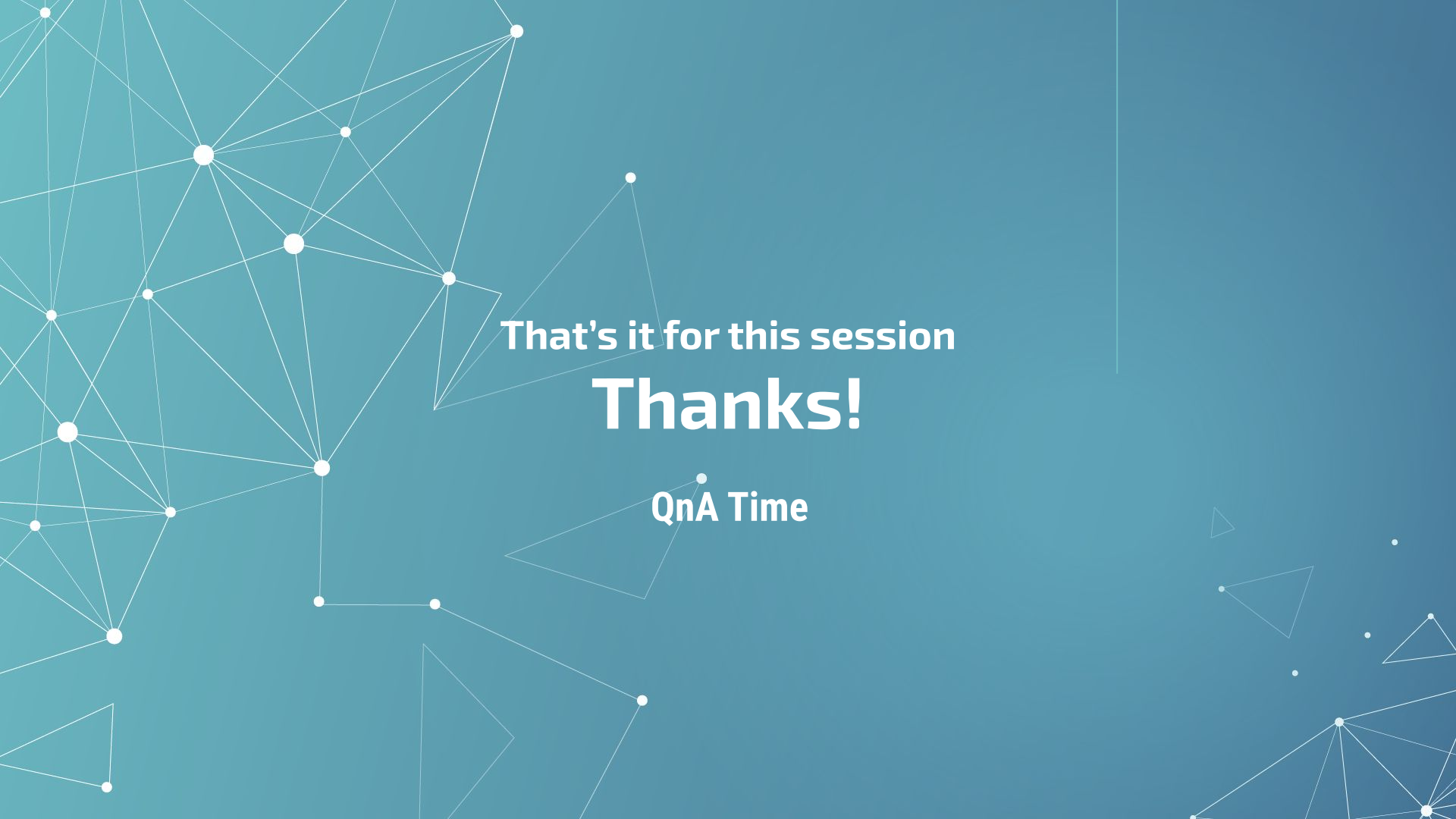
| Variable name    | Description                                           |  |
|------------------|-------------------------------------------------------|--|
| land_area        | size in square miles                                  |  |
| total_population | estimated population in thousands                     |  |
| percent_city     | percent of population in central city/cities          |  |
| percent_senior   | percent of population $\leq 65$ years                 |  |
| physicians       | number of professionally active physicians            |  |
| hospital_beds    | total number of hospital beds                         |  |
| graduates        | percent of adults that finished high school           |  |
| work_force       | number of persons in work force in thousands          |  |
| income           | total income in 1976 in millions of dollars           |  |
| crime_rate       | Ratio of number of serious crimes by total population |  |
| region           | geographic region according to US Census              |  |

# Slide Download Link

You can download these slides from the below link:

<https://docs.google.com/presentation/d/1Khjv0m9A8hUTKM9ZS-iPFFLtHILVVS-MIOSb8fzavISs/edit?usp=sharing>





That's it for this session

**Thanks!**

QnA Time